BMS CAN communication protocol

V1.9

version	date	Created/Modified by	Modification instructions	Reason for modification
V1.0	April 8, 2021		1. First edition	
V1.1	May 27, 2021		1. Instruction to increase the number of query cycles 2. Correction of PF in section 6.3 Value Definition	
V1.2	November 16, 2021		1. Modification of the calculation range for multi packet frame verification 2. Modification of unit description in analog quantity 3. Improve battery type information interest	
V1.3	December 8, 2021		1. Add other SOP values Query Instructions	

V1.4	January 4th, 2022		1. Fixed value query increase CAN_IDX 2. Add fixed value modification index 3. Add Q10 format SOC Read Instruction 4. Default rate changed from 500 kbit/s to 1000 kbit/s 5. Modify the information content after battery alarm 6. Increase charging heartbeat frame	
V1.5	April 22, 2022		1. Add content related to address allocation 2. Improve the description of heartbeat frame data 3. Increase position indicator light order	
V1.6	June 18, 2022		1. Battery alarm information frame Middle, increase pack	

			power	
			Pressure and PCB temperature Description of	
V1.7			1. Add linkage startup command 2. Add linkage shutdown command 3. Add mandatory opening and pre release instructions	
V1.8	November 10, 2023		1. Add Battery Standard Extension Information 1 Command 2. Battery alarm information Increase charging cutoff current	
V1.9	November 12, 2023		1. Explanation of modifying the verification range of multi packet frame data in the "4.3 Transmission Protocol" frame 2. Modify the document format and add Appendix B for explanatory	

			purposes Quasi extended information	

1 Terms and definitions

one point one

Extended Frame

CAN data frames defined in the CAN bus using 29 bit identifiers

one point two

Priority (P)

Set the arbitration priority of the transmission process in a 3-digit domain in the identifier, with the highest priority being level 0

Low priority is at level 7.

one point three

Protocol Data Unit (PDU)

A specific CAN data frame format that includes the CAN ID and data domain.

one point four

Lithium battery BMS

A built-in management system for the battery pack.

one point five

control module

The device in the system that actively communicates with the lithium battery BMS.

2 General Provisions

2.1 The communication network between the lithium battery BMS and other devices adopts the CAN2.0B communication protocol.

2.2 Communication uses extended frame format.

2.3 The data transmission adopts the format of sending low bytes first.

2.4 After more than 20 minutes of not receiving a heartbeat frame from the control module, the lithium battery BMS will stop actively sending it and resume after receiving the heartbeat frame again.

3 physical layer

The physical layer using this protocol shall comply with the provisions of ISO 11898-1:2003 and SAE J1939-11:2006 regarding the physical layer. The communication rate between the control module and the lithium battery BMS should be 1000 kbit/s.

4 data link layer

4.1 Allocation of addresses

When the control module has only one lithium battery BMS in the system, it can ignore the actual address of the lithium battery BMS anddirectly communicate with it using 0x00.

In address allocation, 0x01~0x1F are assignable addresses. 0x80~0xEF are random address ranges.

Table 1 Address allocation range of each device

device	address
Lithium battery BMS	Address range: 0x00~0xEF, 0xFF is used as the broadcast address, and 0x00 is used as any address
control module	Address range: 0xF0~0xFE

4.2 **Protocol Data Unit (PDU)**

Each CAN data frame contains a single protocol data unit (PDU) as shown in Table 2. The protocol data unit consists of seven parts, namely priority (P), reserved bit (R), data page (DP), PDU format (PF), PDU specific (PS), source address (SA), and data domain.

Table 2 Protocol Data Unit (PDU)

P	R	DP	P F	P S	S A	DAT A

	three	one	one	eight	eight	eight	0-64

Explanation: (Data format requirements)

1. P is the priority: set from the highest 0 to the lowest 7.
2. R is the reserved bit: for future development and use, this standard is set to 0.
3. DP is the data page: used to select the auxiliary page for parameter group description, which is set to 0 in this standard.
4. PF is in PDU format: used to determine the format of the PDU and the parameter group number corresponding to the data field.
5. PS is a PDU specific format: in this standard, the PS value is the target address.
6. SA is the source address: the source address where this message is sent.
7. DATA is the data field: If the length of the given parameter array data is ≤ 8 bytes, it is transmitted as 8 bytes, with a default value of 00H. If the given parameter array data length is 9~

1785, data transmission requires multiple CAN data frames and communication through protocol transmission function, as specified in 4.3.

8. The third row of this table represents the number of digits.

4.3 transport protocol

The data frames larger than 8 bytes in this section should be transmitted using the following multi frame transmission protocol.

After receiving multiple frames of protocol data, data verification should be performed. If the verification fails, the transmitted data will be discarded. Multiple frame messages cannot be nested for sending.

For multi frame messages, the message cycle is the entire transmission cycle of the data packet, and the interval time between single frame messages is not less than 10ms.

Table 3 Format of Multi frame Data Transmission Protocol

Frame number	Data1	Data2	Data3	Data4	Data5	Data6	Data7	Data8
one	Current message number	Total number of message frames	Effective data length of message Low Byte	Effective data length of message High Byte	Valid data 01	Valid data 02	Valid data 03	Valid data 04
two	Current message Serial number	Valid data 05	Valid data 06	Valid data 07	Valid data 08			
	Current message Serial number	Valid data N	Check code low byte	Check code high byte	00H	00H	00H	00H

Explanation: (Data format requirements)

1. The effective data length refers to the number of bytes from "effective data 01" to "effective data N".
2. The verification code refers to the cumulative sum from "valid data 01" to "valid data N" (excluding the "current message number" in each frame of the message).
3. The current message sequence number range is: 1-255.

When the last frame is less than 8 bytes, it is transmitted as 8 bytes, and the unused part is set to 00H.

5 packet classification

5.1 General Provisions

This part of the message is divided into command frames, data frames, and heartbeat frames by type.

The length of the message data is 8 bytes. If the actual data is less than 8 bytes, it will be sent as 8 bytes, and the unused part will be set as 00H.

5.2 Command frame

Table 4 Command Frame Classification

Message Description	PF	first power	Data length (Byte)	data type	Message cycle (ms)	Source Address - Destination Address
Identification code query	0x03	six	/	/	Non periodic report writing	Control module - lithium battery BMS
Identification code query should answer	0x04	six	eight	BIN	Non periodic report writing	Lithium battery BMS control module block
Address allocation	0x07	six	eight	BIN	Non periodic report writing	Control module - lithium battery BMS
Position indication	0x0B	six	/	/	Non periodic report writing	Control module - lithium battery BMS
Address allocation response	0x08	six	eight	BIN	Non periodic report writing	Lithium battery BMS control module block
Fixed value query	0x80	six	eight	BIN	Non periodic report writing	Control module - lithium battery BMS
Fixed value query response	0x81	six	eight	BIN	Non periodic report writing	Lithium battery BMS control module block
Cell temperature query	0x82	six	eight	BIN	Non periodic report writing	Control module - lithium battery BMS

Cell temperature query response	0x83	six	eight	BIN	Non periodic report writing	Lithium battery BMS control module block
Individual voltage query	0x84	six	eight	BIN	Non periodic report writing	Control module - lithium battery BMS
Individual voltage query response	0x85	six	eight	BIN	Non periodic report writing	Lithium battery BMS control module block
Cycle count query	0x86	six	eight	BIN	Non periodic report writing	Control module - lithium battery BMS
Cycle count query response	0x87	six	eight	BIN	Non periodic report writing	Lithium battery BMS control module block
Other SOP queries	0x88	six	eight	BIN	Non periodic report writing	Control module - lithium battery BMS
Other SOP queries response	0x89	six	eight	BIN	Non periodic report writing	Lithium battery BMS control module block
Fixed value setting	0xF0	six		BIN	Non periodic report writing	Control module - lithium battery BMS
Fixed value setting response	0xF1	six		BIN	Non periodic report writing	Lithium battery BMS control module block

5.3 Data frame

Table 5 Data Frame Classification

Message Description	PF	first power	Data length (Byte)	data type	Message cycle (ms)	Source Address - Destination Address

Charging request information	0x22	six	eight	BIN	one thousand	Lithium battery BMS control module block
Battery alarm information	0x24	six	eight	BIN	one thousand	Lithium battery BMS control module block
Battery operation information	0x26	six	eight	BIN	one thousand	Lithium battery BMS control module block
Standard Expansion of Battery Meters Exhibition Information 1	0x28	six	N	BIN	one thousand	Lithium battery BMS control module block

5.4 Heartbeat frame

Table 6 Classification of Heartbeat Frames

Message Description	PGN	first power	Data length (Byte)	data type	Message cycle (ms)	Source Address - Destination Address
Heartbeat frame	0x43	six	eight	BIN	two thousand	Control Module - Lithium Battery BMS
Charging Heartbeat Frame	0x45	three	0	/	two point five	Lithium Battery BMS ->Control Module

5.5 Program online update frame

Table 7 Program Online Update Frame Classification

Message Description	PGN	first power	Data length (Byte)	data type	Message cycle (ms)	Source Address - Destination Address
Upgrade Heartbeat Frame	0x70	four	eight	BIN	one thousa	Control module - lithium battery BMS

					nd	
Upgrade Heartbeat Response frame	0x71	four	eight	BIN	one thousand	Lithium Battery BMS - Power control module
Start download command frame	0x72	four	eight	BIN	five hundred	Control module - lithium battery BMS
Start download response frame	0x73	four	eight	BIN	five hundred	Lithium Battery BMS - Power control module
Request interval commands frame	0x74	four	Indefinite	BIN	five hundred	Control module - lithium battery BMS
Request interval response Frame 1	0x75	four	eight	BIN	five hundred	Lithium Battery BMS - Power control module
Request interval response Frame 2	0x76	four	eight	BIN	five hundred	Lithium Battery BMS - Control module
Start packaging command frame	0x77	four	eight	BIN	five hundred	Control module - lithium battery BMS
Start packet answering frame	0x78	four	eight	BIN	five hundred	Lithium Battery BMS - Control module
Data transmission frame	0x79	four	eight	BIN	ten	Control module - lithium battery BMS
Complete the package command	0x7A	four	eight	BIN	five hun	Control module - lithium battery

frame					dre d	BMS
Complete package response frame	0x7 B	four	eight	BIN	five hun dre d	Lithium Battery BMS - Control module
Program verification data frame	0x7 C	four	eight	BIN	five hun dre d	Control module - lithium battery BMS
Program verification response frame	0x7 D	four	eight	BIN	five hun dre d	Lithium Battery BMS - Control module
Immediate reset command frame	0x7 E	four	eight	BIN	five hun dre d	Control module - lithium battery BMS
Immediate reset response frame	0x7 F	four	eight	BIN	five hun dre d	Lithium Battery BMS - Control module

6 Message format and content

6.1 Fixed value query

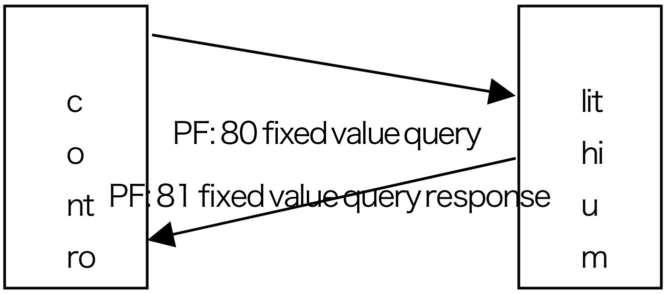


Table 8 Fixed Value Query Command Frame

Starting Byte or Bit	Parameter Name	data format	Field length	Remarks
one	Fixed value serial number	BIN	2Byte	Unit: No resolution: 1/bit Range: 1~200 Offset: 0

Table 9 Fixed value query response frame

Starting Byte or Bit	Parameter Name	data format	Field length	Remarks
one	Fixed value serial number	BIN	2Byte	Unit: No resolution: 1/bit Range: 1~200 Offset: 0

three	Operation return	BIN	1 Byte	Bit7: Success ID 0x00-Failure 0x01-Success Bit6~Bit4: Reserved Bit3~Bit0: Reason for failure 0x00-Success 0x01-No such fixed value 0x02-Read prohibited □ 0x03-Read □ fail 0x04-Prohibit writing 0x05-Given limit exceeded
four	Leave blank	BIN	1 Byte	Fixed to 0
five	Fixed value informatio n	/	/	Refer to Table A.1 in the appendix

6.2 Cell temperature query

When the number of cell temperatures exceeds 8, multi frame transmission is required for transmission.

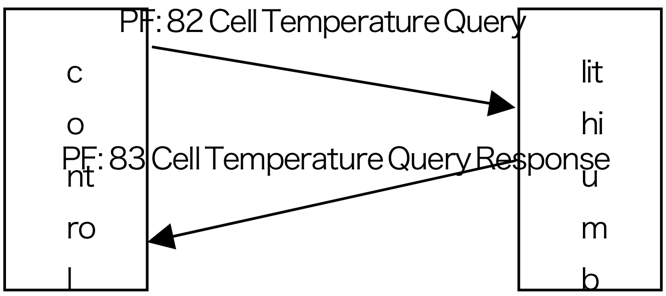


Table 10 Cell Temperature Query Command Frame

Starting Byte or Bit	Parameter Name	data format	Field length	Remarks

Table 11 Cell Temperature Query Response Frame

Starting Byte or Bit	Parameter Name	data format	Field length	Remarks
one	Cell temperature 1	BIN	1 Byte	Unit: ℃ Resolution: 1 ℃ /bit range: 0~160 ℃ Offset: 40
two	Cell temperature 2	BIN	1 Byte	Unit: ℃ Resolution: 1 ℃ /bit range: 0~160 ℃ Offset: 40
three	Cell temperature 3	BIN	1 Byte	Unit: ℃ Resolution: 1 ℃ /bit range: 0~160 ℃ Offset: 40
N	Cell temper	BIN	1 Byte	Unit: ℃ Resolution: 1 ℃ /bit range:

	ature n			0~160℃ Offset: 40

6.3 **Individual voltage query**

When the number of battery cells exceeds 4, multi frame transmission is required for transmission.

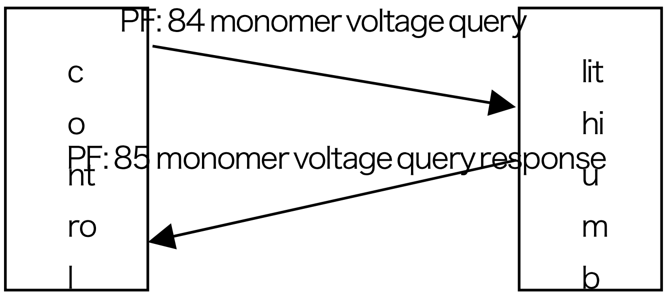


Table 12 Single Unit Voltage Query Command Frame

Starting Byte or Bit	Parameter Name	data format	Field length	Remarks

Table 13 Single Unit Voltage Query Response Frame

Starting Byte or Bit	Parameter Name	data format	Field length	Remarks
one	Cell voltage 1	BIN	2Byte	Unit: V Resolution: 0.001V/bit range: 0~5V Offset: 0
three	Cell voltage 2	BIN	2Byte	Unit: V Resolution: 0.001V/bit range: 0~5V Offset: 0
five	Cell voltage 3	BIN	2Byte	Unit: V Resolution: 0.001V/bit range: 0~5V Offset: 0

N	Cell voltage n	BIN	2Byte	Unit: V Resolution: 0.001V/bit range: 0~5V Offset: 0

6.4 Cycle count query

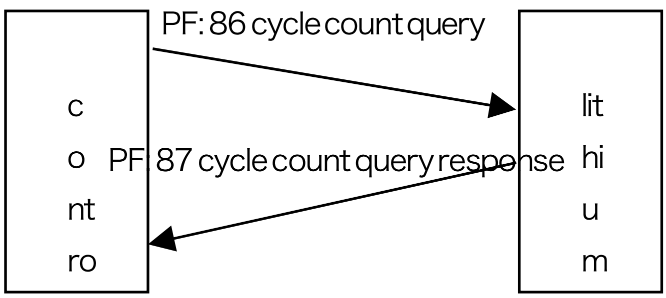


Table 14 Cycle Count Query Command Frame

Starting Byte or Bit	Parameter Name	data format	Field length	Remarks

Table 15 Response Frame for Cycle Count Query

Starting Byte or Bit	Parameter Name	data format	Field length	Remarks
one	Number of cycles	BIN	2Byte	Unit: Times Resolution: 1 time/bit range: 0~65535 Offset: 0

6.5 **Other SOP queries**

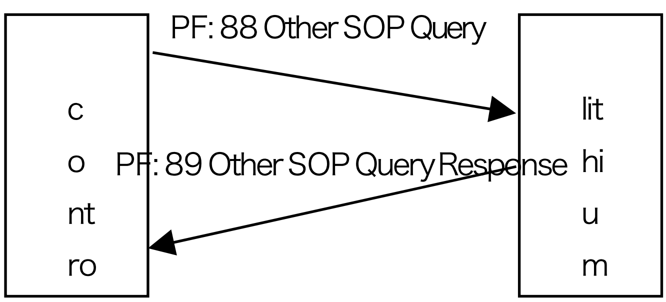


Table 16 Other SOP Query Command Frames

Starting Byte or Bit	Parameter Name	data format	Field length	Remarks

Table 17 Other SOP query response frames

Starting Byte or Bit	Paramete r Name	data format	Field length	Rema rks
one	0.5 second SOP	BIN	2Byte	Unit: W Resolution: 10W/bit range: 0~60kW

				Offset: 0
three	3 second SOP	BIN	2Byte	Unit: W Resolution: 10W/bit range: 0~60kW Offset: 0

6.6 Fixed value setting

Starting Byte or Bit	Parameter Name	data format	Field length	Rema rks
one	Fixed value serial number	BIN	2Byte	Unit: No resolution: 1/bit Range: 1~200 Offset: 0
three	hold	BIN	2Byte	Fixed to 0
five	Fixed value data	/	/	Refer to Table A.1 in the appendix

Starting Byte or Bit	Parameter Name	data format	Field length	Rema rks
one	Fixed value serial number	BIN	2Byte	Unit: No resolution: 1/bit Range: 1~200 Offset: 0
three	Operation return	BIN	1 Byte	Bit7: Success ID 0x00-Failure 0x01-Success Bit6~Bit4: Reserved Bit3~Bit0: Reason for failure 0x00-Success 0x01-No such fixed value 0x02-Read prohibited □ 0x03-Read □ fail 0x04-Prohibit writing 0x05-Given limit exceeded
four	Leave blank	BIN	1 Byte	Fixed to 0

6.7 Charging request information

The charging request information is actively reported by the lithium battery BMS and does not require a response from the main control module.

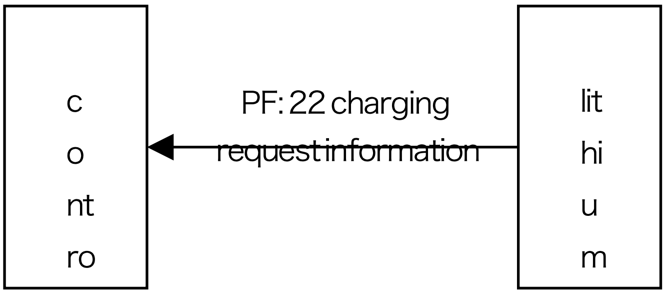


Table 18 Charging Request Information

Starting Byte or Bit	Parameter Name	data format	Field length	Rema rks
one	Request voltage	BIN	2Byte	Unit: V Resolution: 0.01V/bit range: 0~100V Offset: 0
three	Request current	BIN	2Byte	Unit: A Resolution: 0.01A/bit range: 0~600A Offset:0
five	Maximum voltage of single cell	BIN	2Byte	Unit: V Resolution: 0.001V/bit range: 0~5V Offset: 0
seven	Charging state	BIN	2Byte	B15: Temperature limited charging current occurs B14: Number of cycles limit charging current B13: Long term quiescent limit charging current B12: B11: B10: B9: B8: B7: B6: B5: B4: B3: B2: B1: Pre charging required B0: Charging prohibited

6.8 **Battery alarm information**

The battery alarm information is actively reported by the lithium battery BMS and does not require a response from the main control module.

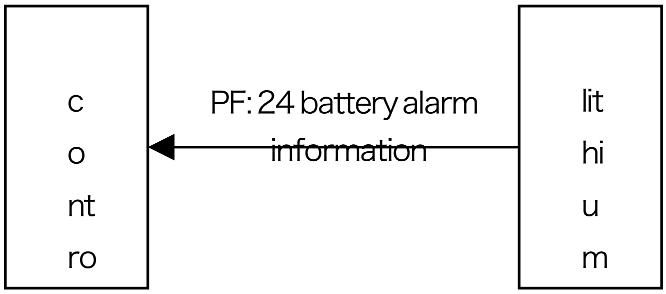


Table 19 Battery Alarm Information

Starting Byte or Bit	Parameter Name	data format	Field length	Remarks
one	Alarm	BIN	2Byte	B15: Discharge overcurrent B14: Battery damage B13: AFE abnormality B12: Low temperature of battery (charging/discharging) B11: High temperature of battery (charging/discharging) B10: Under voltage of battery (single unit) B9: Over voltage of battery (single unit) B8: Temperature sensor abnormal B7: B6: B5: B4: B3: B2: B1:

				B0: Charging overcurrent
three	Warning	BIN	2Byte	B15: Discharge overcurrent B14: B13: B12: Low temperature of battery (charging/discharging) B11: High temperature of battery (charging/discharging) B10: Under voltage of battery (single unit) B9: Over voltage of battery (single unit) B8: B7: B6: B5: B4: B3: B2: B1: B0: Charging overcurrent
five	Pack voltage	BIN	2Byte	Unit: V Resolution: 0.01V/bit range: O ~100V Offset: 0

seven	PCB temperature	BIN	1 Byte	Unit: ℃ Resolution: 1 ℃ /bit range: 0~255 ℃ Offset: 40
eight	Charging cutoff current	BIN	1 Byte	Unit: C Resolution: 0.01C/bit range: 0-2C Offset: 0

6.9 Battery operation information

The battery operation information is actively reported by the lithium battery BMS and does not require a response from the main control module.

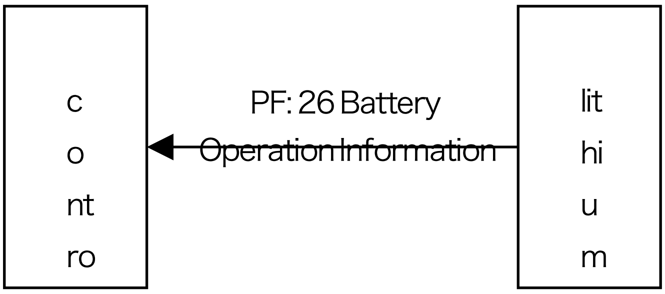


Table 20 Battery
Operation Information

Starting Byte or Bit	Parameter Name	data format	Field length	Remarks
one	Total battery pressure	BIN	2Byte	Unit: V Resolution: 0.01V/bit range: 0~100V Offset: 0
three	Battery current	BIN	2 bytes	Unit: A Resolution: 0.01A/bit Range: -300A~300A Offset: 0
five	SOC	BIN	1 byte	Unit: % Resolution: 1%/bit range: 0~100% Offset: 0
six	SOH	BIN	1 byte	Unit: % Resolution: 1%/bit range: 0~100% Offset: 0

seven	15 second SOP	BIN	2Byte	Unit: W Resolution: 10W/bit range: 0~60kW Offset: 0

6.10 Battery Standard Extension Information 1

Battery standard extension information 1 is a lithium battery BMS active reporting information that does not require a response from the main control module.

Battery standard extension information 1 is customized for customers, please refer to the corresponding supplementary agreement for specific content. If there is no customized content, the lithium battery BMS does not need to be sent.

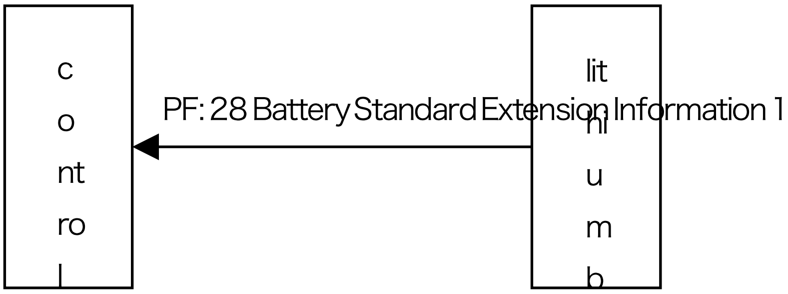


Table 21 Battery Standard Extension Information 1

Starting Byte or Bit	Parameter Name	data format	Field length	Remarks
				Customer customized content, see Table B.1 in Appendix B

6.11 **Heartbeat frame**

After the heartbeat frame is powered on by the main control module, it needs to be periodically sent to the lithium battery BMS, and the lithium battery BMS does not need to respond.

The heartbeat frame is for all devices on the bus, and the PS (target address) must be 0xFF.

Table 22 Heartbeat frames

Starting Byte or Bit	Parameter Name	data format	Field length	Remarks
one	CAN device Pre registration	BIN	4Byte	Registered device's Bitmap Example: If there are currently six addresses of batteries for 1, 2, 3, 4, 5, and 6, the data is 0b000000 000000 000000 01111111

	information			
five	CAN device Registration Information	BIN	4 bytes	

7 **Program online update**
slightly

8 **Address allocation**

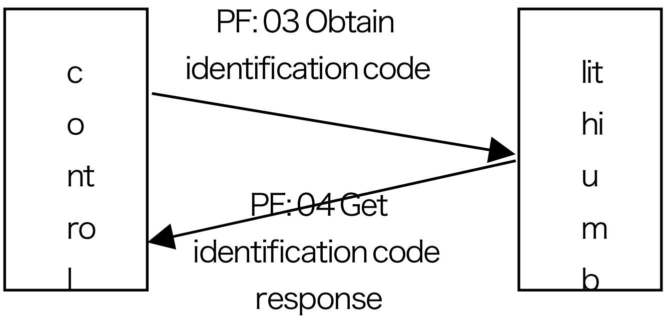


Table 23 Obtaining Identification Codes

Starting Byte or Bit	Parameter Name	data format	Field length	Remarks
one	Obtaining recognition code	/	0Byte	

Table 24 Obtaining Identification Code Response

Starting Byte or Bit	Parameter Name	data format	Field length	Remarks
one	Device identification code	BIN	6Byte	6-byte device unique identification code
seven	Preserve data	BIN	2 bytes	Fixed to 0

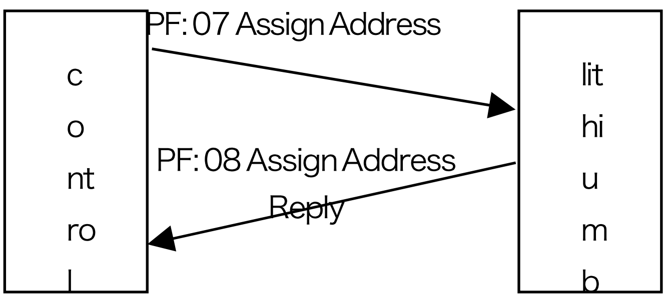


Table 25 Allocation of addresses

Starting Byte or Bit	Parameter Name	data format	Field length	Remarks
one	Device identification code	BIN	6Byte	6-byte device unique identification code
seven	Set Address	BIN	1 byte	Address assigned to the corresponding device
eight	Preserve data	BIN	1 Byte	Fixed to 0

Table 26 Allocation Address Response

Starting Byte or Bit	Parameter Name	data format	Field length	Remarks
one	Device identification code	BIN	6Byte	6-byte device unique identification code
seven	Set Address	BIN	1 byte	Address assigned to the corresponding device
eight	Preserve data	BIN	1 Byte	Fixed to 0

Allocation process:

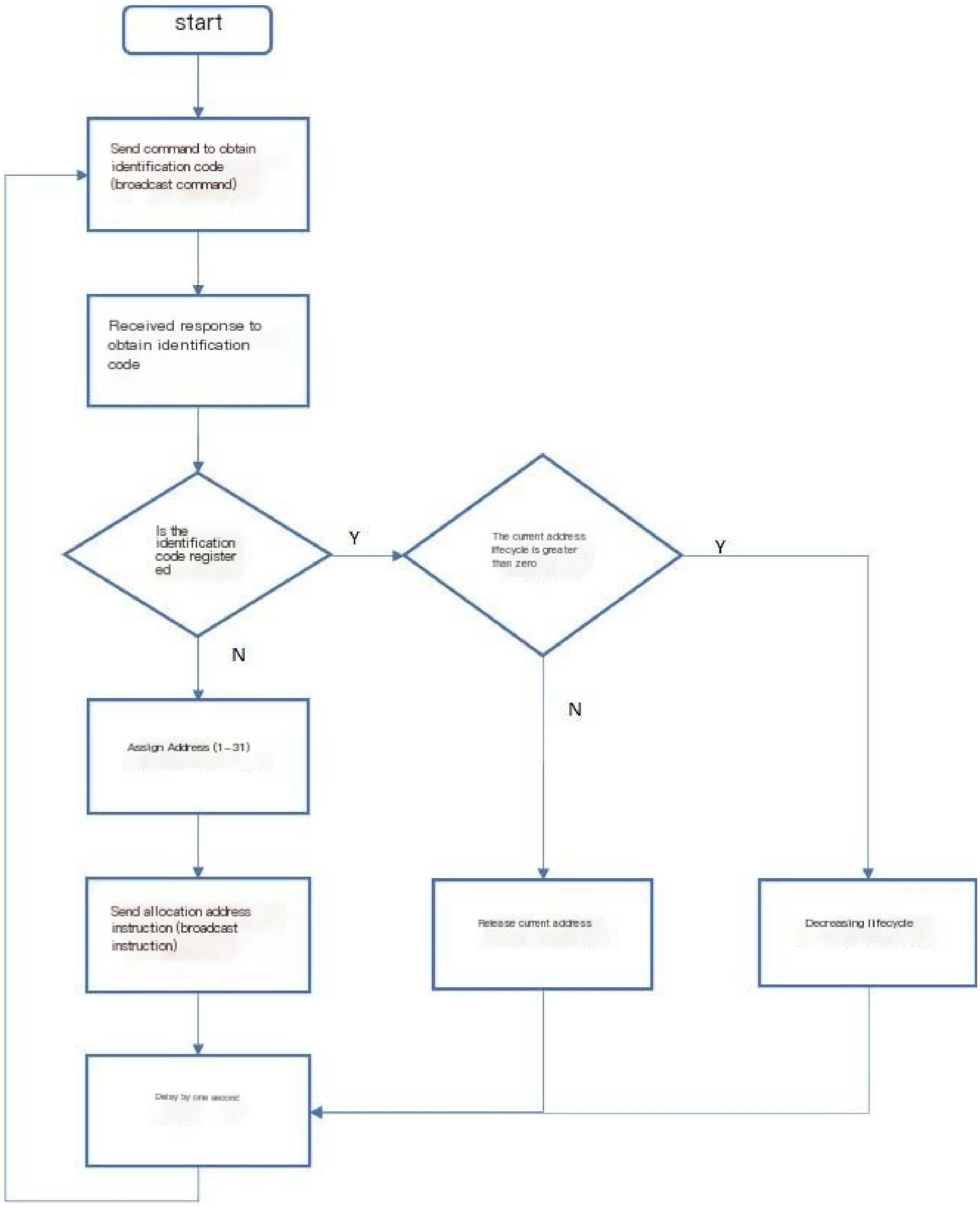
1. The host sends periodic instructions to all devices on the bus to obtain identification codes
2. After receiving the instruction to obtain the identification code, if the device has **not** yet **been** assigned an address, it will immediately randomly generate an address and respond with the local unique identification code
3. After receiving the unique identification code, the host allocates the corresponding address and issues the allocation address instruction
4. After receiving the allocation address instruction, the assigned device records the allocation address and corresponding SA address, and uses them in subsequent communication.

   The default lifecycle of the device address is 15 seconds, and the address lifecycle needs to be refreshed using heartbeat frames. When the device address lifecycle returns to zero, the device will return to the unassigned address state.

   The same host should also set an address lifecycle so that assigned addresses can be released. The host address lifecycle should be greater than 3 seconds and less than the default lifecycle of the device address.

9 **Position indication**

The device corresponding to PS (target address) will periodically flash all LEDs for 10 seconds after receiving this command. Used to indicate the location of the corresponding address battery.


start
Send command to obtain identification code (broadcast command)
Received response to obtain identification code
Is the identification code register ed
Y
N
The current address lifecycle is greater than zero
Y
N
Assign Address (1~31)
Send allocation address instruction (broadcast instruction)
Release current address
Decreasing lifecycle
Delay by one second

Appendix A
(Normative
Appendix) Lithium
Battery BMS Setting
Information Table

Please refer to Table A.1 for the BMS setting information of lithium batteries.

Table A.1 Lithium Battery Setting Information Table

Fixed value serial number	Parameter Name	data format	Field length	Remarks
one	Equipment model	ASCII code	32Byte	Read Only
two	Reserved	BIN	2Byte	
three	Equipmen t serial number	ASCII code	32Byte	Read Only
four	Device controlle r hardware version	BIN	2Byte	Read Only The hardware version number consists of: major version number and minor version number. Example: The hardware version number is V1.00, the major version number is identified as 01H, and the minor version number is identified as 00H; The sending order of message bytes is 01H 00H.
five	Device controlle r software version	BIN	2Byte	Read Only The software version number consists of: major version number and minor version number. Example: The software version number is V1.00, the major version number is identified as 01H, the minor version number is identified as 00H, and the message byte sending order is 01H 00H.
six	Device controlle r	BIN	4Byte	Read only item Data1: Year Data2: Month Data3: Date Data4:

	software date			Hour Example: The software date is 18:00 on April 8, 2021, indicating 21H 04H 08H 18H, and the message byte sending order is 21H 04H 08H 18H.
seven	CAN protocol version	BIN	2Byte	Read Only The composition of the CAN protocol version number is divided into: major version number and minor version number. Example: The CAN protocol version number is V1.00, the major version number is identified as 01H, the minor version number is identified as 00H, and the message byte sending order is 01H 00H.
eight	Number of battery cells	BIN	1 Byte	Unit: pieces Resolution: 1/bit Range: 1-16 Offset: 0
nine	Cell type	BIN	1 Byte	0: Lithium iron phosphate 1: Lithium cobalt oxide 2: Other ternary lithium batteries 3: Solid state lithium battery
ten	Cell temperature transmission	BIN	1 Byte	Unit: pieces
	Number of sensors			Resolution: 1/bit Range: 1-255 Offset: 0
eleven	Number of ambient temperat	BIN	1 Byte	Unit: pieces Resolution: 1/bit Range: 0-255

	ure sensors			Offset: 0
twelve	Number of other temperature sensors	BIN	1 Byte	Unit: pieces Resolution: 1/bit Range: 0-255 Offset: 0
thirteen	Reserved	BIN	2Byte	
fourteen	PACK rated voltage	BIN	2Byte	Unit: V Resolution: 0.01V/bit range: O ~100V Offset: 0
fifteen	PACK rated capacity	BIN	2Byte	Unit: AH Resolution: 0.01AH/bit Range: 0~300AH Offset: 0
sixteen	Maximum discharge current	BIN	2Byte	Unit: A Resolution: 0.01A/bit range: 0~600A Offset:0
seventeen	Maximum charging current	BIN	2Byte	Unit: A Resolution: 0.01A/bit range: 0~600A Offset:0
eighteen	FOV voltage			
nineteen	OVP voltage			

twenty	Platform voltage			
twenty-one	Reserved			

Appendix B
Battery Standard Extension Information 1

Table B.1 Standard Extension Information 1

twenty-one